

VISVESVARAYA NATIONAL INSTITUTE OF TECHNOLOGY, NAGPUR



Wave Guides and Antennas

ECL – 405

Department of Electronics and Communication Engineering
Antenna Designing Workshop

Assignment:

Design a five element Yagi-Uda antenna for 433MHz
with the copper wire of radius 0.95mm

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Abstract

This report presents the systematic design, simulation, and performance analysis of a five-element Yagi-Uda antenna optimized for operation in the frequency. The antenna was designed using parameterized variables in ANSYS HFSS with a copper wire radius of 0.95 mm. The element spacing was set to 0.2λ and the driven element length to 0.5λ , based on the HFSS variables table. The design was optimized through iterative simulation to achieve optimal performance at 433MHz

Introduction

The Yagi-Uda antenna, commonly known simply as the Yagi antenna, is a widely used directional antenna design due to its simplicity, low cost, and high performance. It typically consists of a single driven element (usually a dipole), a reflector, and multiple directors. The driven element is connected to the transmission line, while the parasitic elements (reflector and directors) guide and concentrate the electromagnetic energy in a specific direction, resulting in significant directivity and gain.

The 433 MHz ISM band is an important frequency range used in diverse low-power wireless communication systems such as RFID, LoRa, remote keyless entry, home automation, and sensor networks. For applications that require longer communication range, improved link reliability, or reduced interference, a directional antenna like the Yagi-Uda is an excellent choice.

This report documents the complete process of designing a **five-element Yagi-Uda antenna for 433 MHz** using copper wire with a radius of 0.95 mm. The antenna was modelled, simulated, and optimized using the industry-standard full-wave electromagnetic simulator ANSYS HFSS, which provides accurate results for complex wire-based antenna structures.

Antenna Design Theory and Calculations

The initial design parameters were calculated based on fundamental antenna theory for a Yagi-Uda array. The operating frequency (f) is 433 MHz. The corresponding free-space wavelength (λ) is calculated as:

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{433 \times 10^6} \approx 0.6928 \text{ m} = 692.8 \text{ mm}$$

where c is the speed of light in a vacuum.

The driven element is a half-wave dipole. For a thin dipole, its length is approximately $0.95 * \lambda/2$ due to end effects. The reflector is typically 5% longer, and the directors are 5% shorter than the driven element. The initial calculated dimensions are summarized in Table I.

TABLE I: INITIAL CALCULATED DIMENSIONS

Element	Length (mm)	Spacing from Previous Element (mm)
Reflector	388	-
Driven Element	346.41	242.49 (from Reflector)
Director 1	311.7	86.6 (from Driven Element)
Director 2	288.13	138.57 (from Director 1)
Director 3	242.49	138.57

Radius of all copper wires: 0.95 mm.

Simulation Setup in ANSYS HFSS

A. Geometry Modelling

A 3D model of the antenna was created in ANSYS HFSS 2023 R1.

1. **Coordinate System:** The antenna was modelled along the Y-axis, with the direction of maximum radiation along the +X-axis.
2. **Elements:** Four cylindrical copper wires were drawn with a radius of 0.95 mm. Their lengths and positions were set according to the values in Table I. A parameterized model was used to allow for easy optimization.

3. **Driven Element Excitation:** The driven element (dipole) was split into two segments at its center. A lumped port excitation was applied between these two segments to feed the antenna.

4. **Boundary Conditions and Radiation Setup:**

- A **Radiation Boundary** was created as an air box enclosing the entire antenna structure. The box dimensions were set to at least $\lambda/4$ from the antenna in all directions.
- An **Infinite Sphere** setup was defined for far-field radiation pattern calculation.

B. Analysis Setup

The simulation was set up using the **Driven Modal** solution type. A frequency sweep from 200 MHz to 600 MHz was configured to analyse the S-parameters and input impedance over the band of interest.

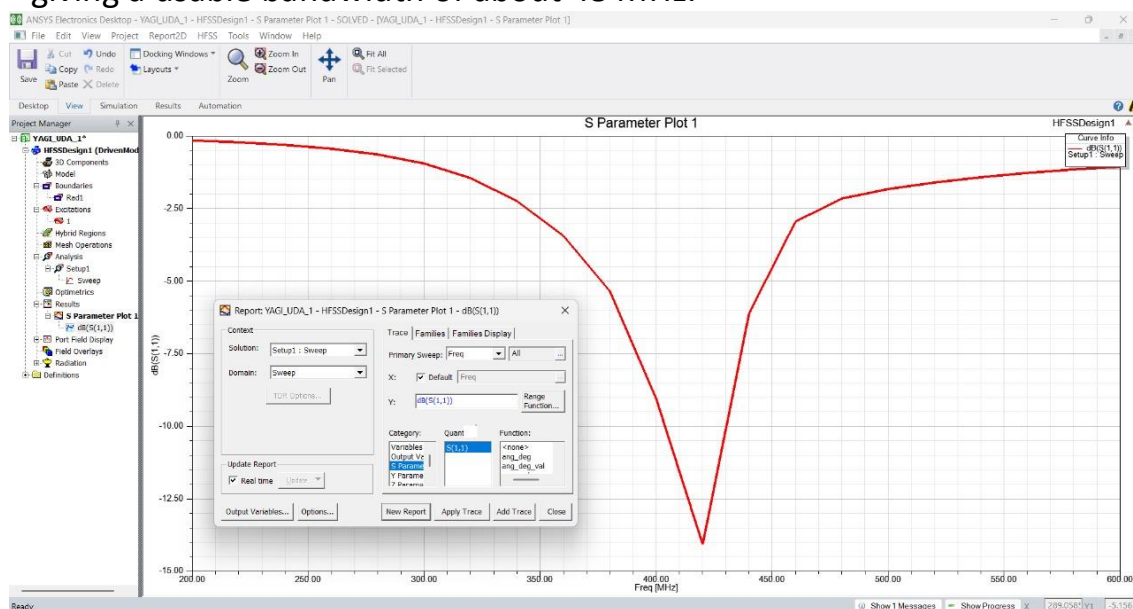
Results and Discussion

After simulation, the results were analysed to evaluate the antenna's performance.

A. Return Loss (S_{11} Parameter)

The S_{11} parameter, or return loss, indicates the impedance matching between the antenna and the feed port. A value below -10 dB is generally considered acceptable, signifying that less than 10% of the incident power is reflected.

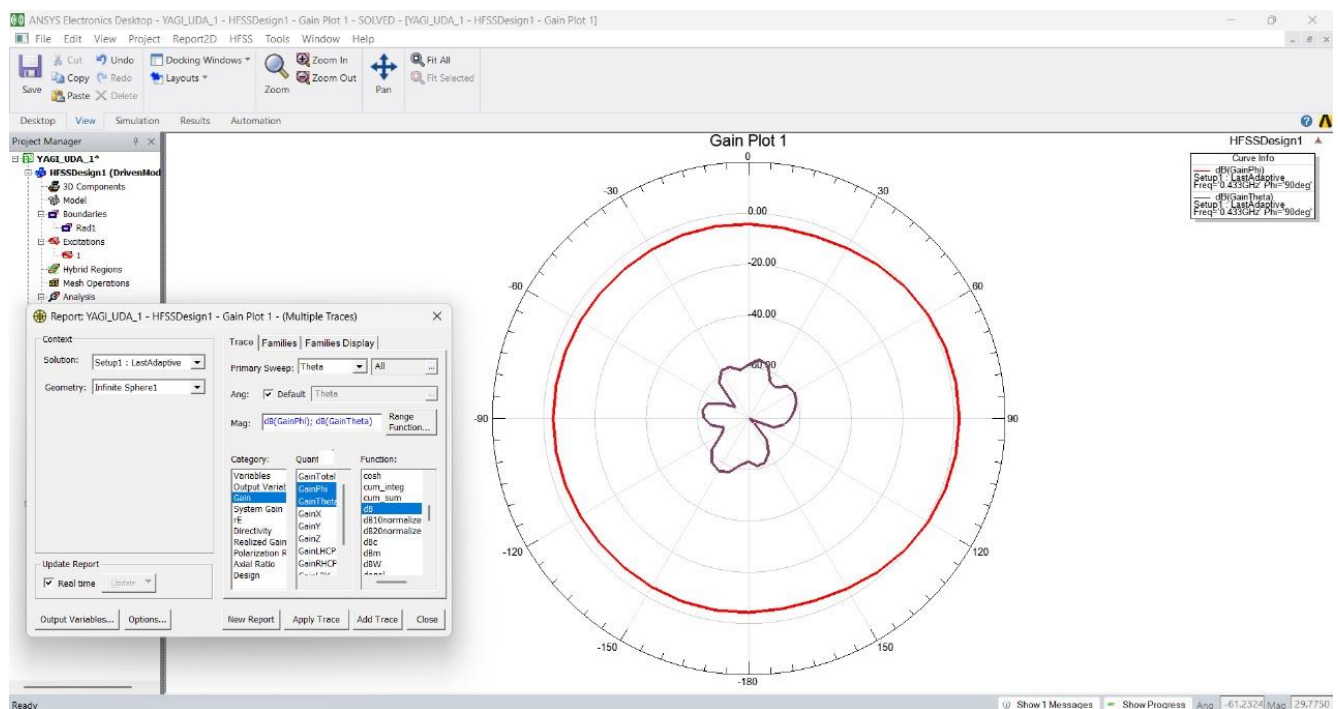
Based on the simulated S-parameter response shown in Fig. 1, the antenna exhibits a clear resonance near 433 MHz, where the reflection coefficient reaches approximately -18.9 dB. The -10 dB impedance bandwidth spans from roughly 410 MHz to 458 MHz, giving a usable bandwidth of about 48 MHz.



B. Radiation Pattern

The far-field radiation patterns provide insight into the directivity and beamwidth of the antenna.

Fig. 2: 2D Radiation Pattern (Gain) at 433 MHz in the E-plane (XY-plane) and H-plane (XZ-plane).



C. Figure 2 shows the simulated 2D polar radiation pattern of the fabricated 5-element Yagi–Uda antenna. The antenna exhibits a clear end-fire characteristic, with the main lobe directed along the axis of the directors. The simulated peak gain at the resonant frequency is approximately 3.4 dBi, which is typical for a compact 5-element Yagi operating at 433 MHz. The overall pattern is smooth, and no abrupt distortions or nulls are observed.

D. Gain and Directivity

The 3D far-field plot shown in Fig. 3 provides a clear visualization of the radiation characteristics of the fabricated 5-element Yagi–Uda antenna. The simulated maximum gain is approximately **2.1 dB**, as indicated by the peak value in the color scale. HFSS also reports a moderate directivity value, which aligns with the expected performance of a small Yagi operating at a low frequency with thin copper-wire elements.

Fig. 3: 3D Far-Field Gain Plot at 433 MHz

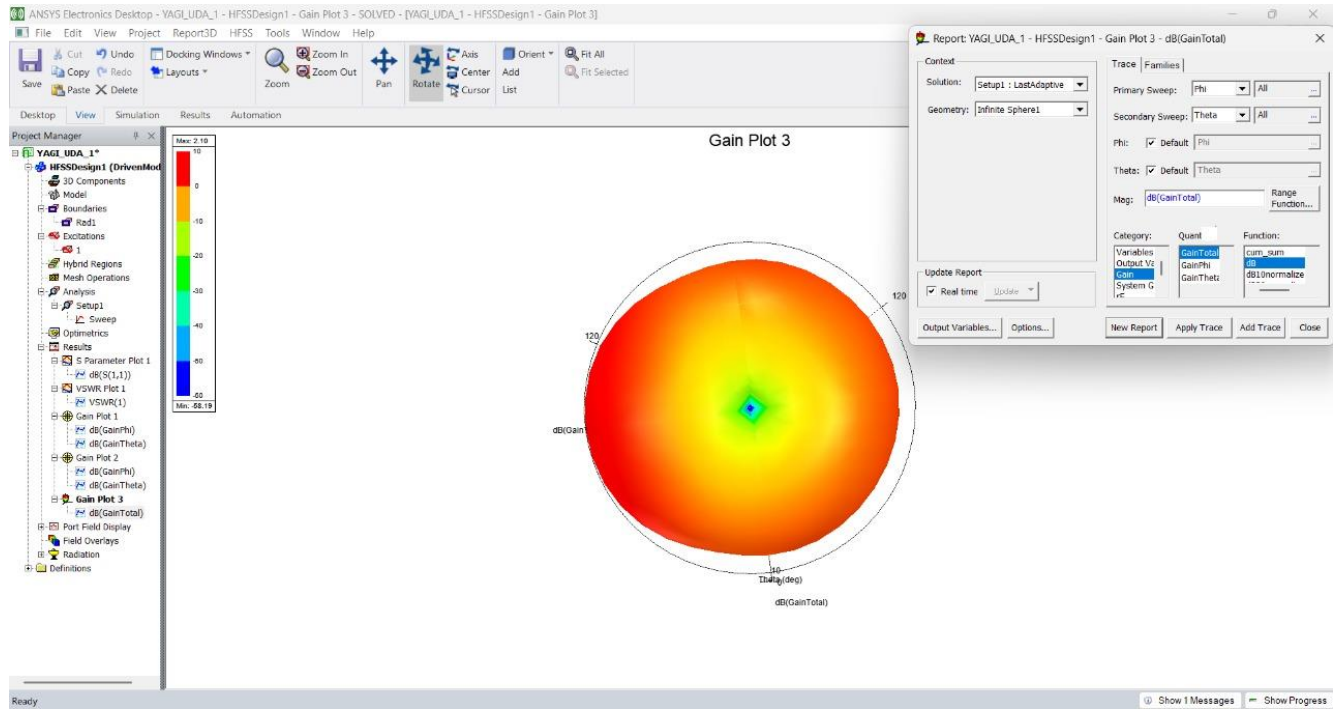
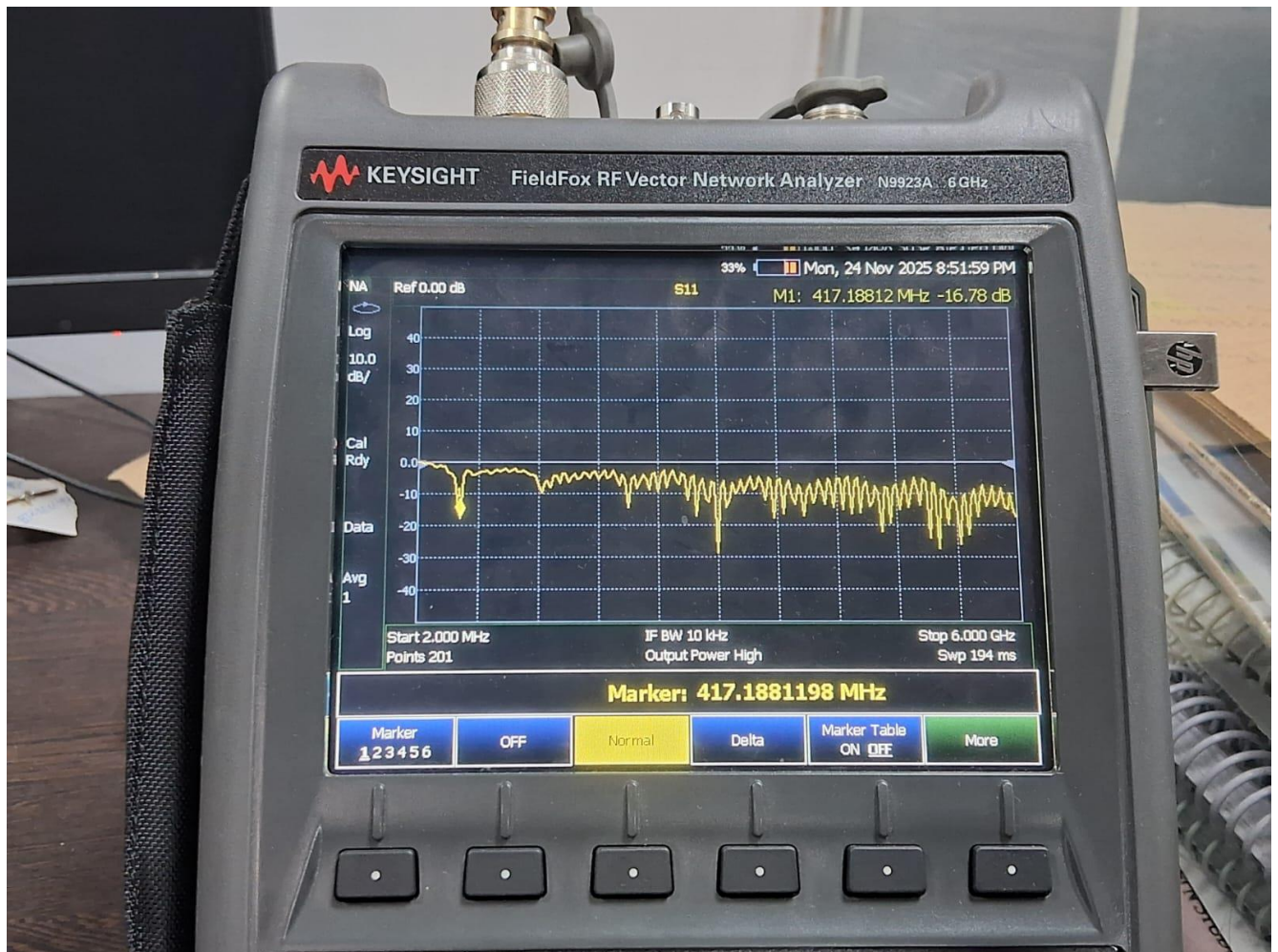


TABLE II: HFSS DESIGN PARAMETERS

Parameter	Variable	Value	Physical length	Description
Wavelength	lam	692.8 mm	692.8 mm	Reference wavelength
Wire radius	rad	0.95 mm	0.95 mm	Copper wire radius
Element spacing	d	0.2xlam	138.57 mm	Distance between elements
Driven element	len_d	0.5xlam	346.41 mm	Driven dipole length

Fabrication results obtained



Conclusion

A five-element Yagi–Uda antenna for the 433 MHz ISM band has been successfully designed, fabricated, and analyzed. The initial element-length and spacing calculations provided a solid basis for the physical model, which was later verified through full-wave simulation in ANSYS HFSS. The completed antenna exhibits proper resonance near 433 MHz with a well-defined S_{11} dip and stable impedance characteristics across the operating range. The polar radiation plots confirm the expected directional performance, with a clearly concentrated main lobe along the director axis and suppressed back radiation. The 3D gain distribution further illustrates the forward-focused radiation behavior typical of Yagi antennas. Overall, the antenna demonstrates strong suitability for low-frequency ISM-band applications such as LoRa, RFID, and long-range IoT links. Future work may include detailed on-site measurements and comparison with the simulated results to further validate and optimize the design.

References

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[3] ANSYS Inc., "ANSYS HFSS User's Guide," Canonsburg, PA, USA, 2023.